

Project

Document role

- This file owns project scope, fixed constraints, milestone framing, sequencing posture, and near-term priority calls.
- Keep detailed subsystem design in `docs/core/SYSTEMS.md` and final 48V electrical architecture in `docs/core/ELECTRICAL_48V_ARCHITECTURE.md`.
- Keep decision/risk/open-question state in `docs/core/TRACKING.md`, not as a second decision log here.
- Keep exact topology, fuse, conductor, and bench procedure detail in `docs/implementation/`.

Snapshot

- As-of date: 2026-06-01
- Phase: post-install interior buildout and measured systems-layout validation
- Install milestone: May 7, 2026 Hiatus install is now historical context; the real camper shell is available for physical mockups and measurements
- Mission: build a reliable off-grid camper workspace suitable for full-time professional use

Fixed constraints

- Truck platform is fixed: 2021 F-350 Regular Cab Long Bed
- Existing truck modifications are fixed: 2.5" lift, 37" tires, 4.88 regear
- Camper platform is fixed: installed/in-hand Hiatus shell on the F-350 long bed

Objectives

- Support standard workdays without power anxiety
- Maintain reliable internet for meetings and high-value workflow tasks
- Provide ergonomic, repeatable workstation setup
- Keep system serviceable and safe in multi-climate travel use

Current execution focus (post-install interior buildout)

- Use [PROJECT_build_order_of_operations](#) as the active phase sequence. Use [INSTALL_MINUS_12_READINESS_PLAN](#) and [STARTER_PLAN_electrical_and_flooring_pre_camper](#) as historical install-window/pre-camper references unless they are explicitly refreshed.
- Use [INTERIOR_furniture_layout_and_galley](#) as the current owner for the office-first passenger-side lofted fridge/wet-spine concept, separated battery bench, and 10-series 80/20 overlay-panel direction.
- Use [ELECTRICAL_48V_ARCHITECTURE](#) as the canonical 48V design reference while mechanically laying out the electrical module and alternator-control path.
- Active work now includes measured camper block envelopes, electrical module live-commissioning cleanup, fuse/device access checks, 80/20 prototype stock/hardware planning, camper audio placement/routing validation, floor/subfloor gates, and payload/weight tracking.
- Electrical status has moved past dead-layout only: the 48V bus, SmartShunt, Orion, Cerbo, MultiPlus inverter mode, and a short limited-current shore-charge test have all been proven without observed errors.
- Defer irreversible work (final shell penetrations, shell-dependent final cable cuts, final extrusion vendor cuts, permanent skins, and finish-floor glue-down) until physical measurement, service-map, access, and post-energization checks pass.

Scope

- In scope: electrical, charging, solar, communications, interior structure/cabinetry, operations process, maintenance tracking
- Out of scope for phase 1: non-critical aesthetic upgrades and major unrelated truck mechanical projects

Requirements baseline (draft)

- Workday autonomy target: modeled profiles (BOM loads + owner-supplied office loads + moderate camper-audio use) indicate 3,794 Wh/day (`core_workday`) and 4,587 Wh/day (`winter_workday`); battery reserve floor policy is 20% minimum SOC
- Connectivity reliability target: Starlink is planned primary path; fallback strategy still TBD
- Thermal comfort and electronics protection limits: TBD
- System safety requirements: fusing, disconnects, wire protection, labeling, service access

Workbook intake (from `Camper Build.xlsx` , imported 2026-02-11)

- Current camper config notes indicate Hiatus base model with vertical lower frame, double back doors, additional windows, Maxxair fan, and overhead LEDs.
- Consult note indicates "Into production 3rd week of February" and that delivery timeline adherence was reported as strong.
- Cabover reference captured: approximately 41 in .
- Truck config in workbook matches fixed constants: 2021 F-350, 8 ft bed.

Scope clarifications from workbook

- In scope for Phase 1 includes office-first electrical architecture, Starlink support, cabinetry/workstation, and core plumbing.
- Truck mechanical upgrades listed as "skip for now" remain out of scope for Phase 1.
- Several passthrough decisions (solar and diesel/propane routing) are still pending and tracked in docs/core/TRACKING.md ; shore hardware is selected, with inlet/cable-support details handled during physical install.

Milestone plan (draft)

- M0: requirements baseline frozen
- M1: system architecture selected (electrical + comms + thermal)
- M2: BOM v1 complete and procurement started
- M3: camper installed / physical shell available for real measurements
- M4: subsystem bench validation complete
- M5: measured interior block envelopes and service map complete
- M6: electrical module first-live checkpoint complete; hard-mounting, procurement, charger programming, and strain-relief closeout underway
- M7: shakedown and punch-list closure

Build sequencing baseline (as-of 2026-05-27)

- Build sequencing is tracked as an overlapping phase plan (not strict serial trades).
- May 7 install readiness is historical context; current sequencing focus is measured interior envelopes, post-energization electrical cleanup/programming, targeted prototype extrusion/hardware ordering, and staged AC-out/alternator validation.
- CAD and AI-generated diagrams are planning references only; real camper measurements and physical mockups own final dimensions.
- Detailed stage-by-stage order of operations, dependencies, and draft date windows:
- [PROJECT_build_order_of_operations](#)

Immediate next decisions

- **Measured envelopes:** physically tape/cardboard the passenger-side lofted fridge/wet-spine skeleton, pump/accumulator board, water tank overlap, separated battery bench, driver-side electrical box/step box, 24 in x 48 in wheel-well desk, DC shelf, diesel heater zone, and workstation before final cut lists.
- **Extrusion/furniture:** order/use 80/20 as controlled prototype stock/hardware, not a final vendor-cut enclosure. Default toward 10-series unless real point loads or restraint needs prove otherwise.
- **Panel strategy:** use overlay/removable panels on living-facing surfaces, keep service zones exposed or quick-removable, and use magnets only for light service covers with locator/backup retention where panel loss matters.
- **Electrical:** current live-commissioning checkpoint is complete: owner confirmed 55.5V throughout the 48V system including at the MultiPlus, MultiPlus inverter mode came on with normal light/hum and no errors, SmartShunt and Orion are visible in VictronConnect, Cerbo GX access point/remote-console workflow is active, and a short shore-charge test ran at limited household-outlet current.
- Purchased Phase 1 AC baseline remains shore source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> single 6-way AC DIN enclosure -> 30A AC-in breaker/disconnect -> MultiPlus AC-in , with 30A AC-out main plus two 20A GFCI branches for later AC-out validation. SKU lock is recorded in BOM rows 13 , 14 , 15 , 41 , 107 , 108 , 109 , 110 , 113 , 114 , 123 , 179 , and 180 ; cable glands/clamps/grommets are on-hand fitment stock.
- AC-in and AC-out may share the purchased single 6-way enclosure only if neutral paths remain isolated, PE/equipment grounding is continuous, no fixed downstream neutral-ground bond is added, and labeling/barriers/strain relief are verified during install. AC-out branch validation is still pending.
- Before sustained charging, program/verify the MultiPlus LiFePO4 charge profile via MK3-USB + VEConfigure or equivalent: absorption/float per Dumfume manual, equalization off, lithium temperature-compensation behavior confirmed, charger current conservative, and DVCC left disabled unless a documented BMS/GX control path is later added.
- Defer Mechman alternator installation until shore charging/monitoring are fully programmed/proven and alternator vendor/support gates are closed.
- **Flooring gate:** do not glue Lonseal or permanently close the floor until EPS/subfloor fit, moisture path, hardpoint, and penetration/service-access checks pass in the installed camper.

- **Payload gate:** capture door-sticker payload and staged scale weights. Treat total payload, rear axle load, rear tire load, and left/right balance as active constraints.

Post-install build restart sequence

Purpose: turn the installed camper shell into measured block envelopes before buying hard-to-return interior structure.

1. Shakedown / inspect

- Stay local enough to address early Hiatus issues if practical.
- Check barn doors, cap seals, bed rail caps, windows/fan, dust paths, rattles, and water paths.

2. Measure / map

- Capture interior dimensions, hardpoints, bed rail interface, floor clearances, door swing, window/fan locations, and candidate pass-through zones.
- Mark no-drill/no-load zones and likely service chases.

3. Floor closeout

- Visually inspect sealed bed paths after cure and normal travel/shakedown; look for lifted tape, cracked beads, dust trails, or dampness.
- Trim EPS flat in ribs, verify plywood flushness, and keep hardpoint pockets non-compressible.
- Glue Lonseal only after the gate passes.

4. Block envelopes before extrusion

- Tape/cardboard the passenger-side lofted fridge/wet-spine skeleton, purchased 36 gal wheel-well tank, under-fridge pump/accumulator board, separated battery bench, driver-side electrical closet/DC shelf, diesel heater base zone, and desk.
- Use the interior furniture/galley concept doc as the layout checkpoint, then verify against real roof-down sweep, cabover/bench cushion use, fridge lid/vent/cord envelope, tank overlap/fitting orientation, pump service access, battery extraction, and wet/dry separation.
- Default to 10-series 80/20, plywood, panels, brackets, and light rail unless measured heavy/dynamic modules prove they need larger extrusion.

5. Electrical bench path

- Continue from the completed first live checkpoint: verify torque/witness marks, covers, labeling, SmartShunt/Cerbo power-source behavior, and shutdown/de-energization behavior.
- Program the MultiPlus battery charge profile before sustained charging, then repeat shore-charge validation with logged voltage/current/SOC/temperature.
- Keep AC-out branch/GFCI validation and alternator commissioning as separate later gates.

Cabinetry approach (recommended)

- Treat 80/20 as the *structure* and simple panels as the *skins*.
- Prioritize serviceability over concealment: every fuse/disconnect should be reachable without uninstalling major devices.
- Build “functional first”: once the electrical module works and survives vibration, add frosted panels + LED backlighting as a Phase 2 overlay.